

```

data <- iris

percent <- as.numeric(
  readline("Enter training percentage: ")
)

set.seed(10)

n <- nrow(data)

train_n <- round(n * percent / 100)

index <- sample(1:n, train_n)

train <- data[index, ]
test <- data[-index, ]

# Logistic model for Setosa
m1 <- glm(
  I(Species == "setosa") ~ Petal.Length + Petal.Width,
  data = train,
  family = binomial
)

# Logistic model for Versicolor
m2 <- glm(
  I(Species == "versicolor") ~ Petal.Length + Petal.Width,
  data = train,
  family = binomial
)

# Logistic model for Virginica
m3 <- glm(
  I(Species == "virginica") ~ Petal.Length + Petal.Width,
  data = train,

```

```
family = binomial
)
# Get probabilities
p1 <- predict(m1, test, type = "response")
p2 <- predict(m2, test, type = "response")
p3 <- predict(m3, test, type = "response")
# Select highest probability
result <- cbind(p1, p2, p3)
prediction <- apply(
  result,
  1,
  which.max
)
```

```
prediction <- c(
  "setosa",
  "versicolor",
  "virginica"
)[prediction]
```

```
# Confusion matrix
```

```
cm <- table(
  Actual = test$Species,
  Predicted = prediction
)
```

```
cat("\nConfusion Matrix:\n")
```

```
print(cm)
```

```

# Accuracy

accuracy <- mean(

  prediction == test$Species

)

cat("\nAccuracy:",

    round(accuracy * 100, 2),

    "%")

```

The screenshot shows the RStudio interface with the following components:

- Script Editor:** Contains R code for calculating accuracy and printing a confusion matrix. The code is as follows:


```

54 "setosa",
55 "versicolor",
56 "virginica"
57 ) [prediction]
58
59 # Confusion matrix
60 cm <- table(
61   Actual = test$Species,
62   Predicted = prediction
63 )
64
65 cat("\nConfusion Matrix:\n")
66 print(cm)
67
68 # Accuracy
69 accuracy <- mean(
70   prediction == test$Species
71 )
72
73 cat("\nAccuracy:",
74     round(accuracy * 100, 2),
75     "%")

```
- Environment:** Shows the 'train' object with 135 observations and 5 variables. The variables are:
 - accuracy: 0.866666666666667
 - cm: 'table' int [1:3, 1:3] 7 0 0 3 0 0 2 3
 - index: int [1:135] 137 74 112 72 88 15 143 149 24 13 ...
 - n: 150L
 - p1: Named num [1:15] 1 1 1 1 1 ...
 - p2: Named num [1:15] 0.227 0.227 0.253 0.2 0.256 ...
 - p3: Named num [1:15] 2.22e-16 2.22e-16 2.22e-16 2.22e-16 2.22e-16 ...
 - percent: 90
 - prediction: chr [1:15] "setosa" "setosa" "setosa" "setosa" "setosa" "setosa" "setosa" "versicolor" ...
 - train_n: 135
- Console:** Shows the output of the R code:


```

> source("C:/Users/2021e/Desktop/EX 1.R")
Enter training percentage: 90

Confusion Matrix:
      Predicted
Actual   setosa versicolor virginica
setosa    7         0         0
versicolor 0         3         2
virginica  0         0         3

Accuracy: 86.67 %

Warning messages:
1: glm.fit: algorithm did not converge
2: glm.fit: fitted probabilities numerically 0 or 1 occurred
3: glm.fit: fitted probabilities numerically 0 or 1 occurred

```